



Optimizing Oil Production through Agile Project Execution Frameworks in Complex Energy Sector Challenges

Magnus Okechukwu Kanu ^{1*}, Ikiomoworio Nicholas Dienagha ², Wags Numoipiri Digitemie ³, Elemele Ogu ⁴, Peter Ifechukwude Egbumokei ⁵

¹ Shell Petroleum Development Company of Nigeria Ltd, Nigeria

² Shell Petroleum Development Company, Lagos Nigeria, Nigeria

³ Shell Energy Nigeria PLC, Nigeria

⁴ Total Energies Exploration & Production Nigeria Limited, Nigeria ⁵ Shell Nigeria Gas (SEN/ SNG), Nigeria

* Corresponding Author: **Magnus Okechukwu Kanu**

Article Info

ISSN (online): 2582-7138

Volume: 03

Issue: 01

January-February 2022

Received: 17-12-2021

Accepted: 15-01-2022

Page No: 769-775

Abstract

The complexity and dynamic nature of the oil production industry necessitates continuous optimization of production processes to maintain operational efficiency and competitiveness. This paper explores the application of Agile project execution frameworks to address the multifaceted challenges within the energy sector. Agile principles, characterized by flexibility, iterative development, and enhanced team collaboration, offer a promising alternative to traditional project management approaches. The paper identifies key industry challenges, analyzes the benefits of Agile methodologies, and outlines strategies for successful implementation. Oil production projects can achieve greater adaptability, improved communication, and accelerated problem-solving capabilities by integrating Agile frameworks. The paper concludes with actionable recommendations for industry stakeholders on adopting Agile practices to drive innovation and efficiency in oil production.

DOI: <https://doi.org/10.54660/IJMRGE.2022.3.1.769-775>

Keywords: Agile project execution, Oil production optimization, Energy sector challenges, Iterative development, Team collaboration, Operational efficiency

1. Introduction

The oil production industry is a cornerstone of the global economy, providing essential energy resources that fuel transportation, power industries, and heat homes. However, this sector faces numerous complexities that can hinder efficiency and productivity (Gielen *et al*, 2019). These complexities arise from various factors including fluctuating market prices, regulatory constraints, geopolitical tensions, and the intrinsic technical challenges of extracting oil from increasingly difficult environments. The need for continuous innovation and optimization in production processes is paramount to maintaining competitiveness and ensuring sustainable operations (Pelluru, 2022).

Optimizing production processes in the oil industry is crucial for several reasons. Firstly, efficient production processes can significantly reduce operational costs, thus enhancing profitability. Given the volatile nature of oil prices, maintaining costeffectiveness is essential for the financial health of oil companies (Arya, 2022). Secondly, optimized processes contribute

to improved safety and environmental performance. Companies can comply with stringent environmental regulations and reduce their ecological footprint by minimizing waste and emissions. Thirdly, in the face of depleting easy-to-access oil reserves, the industry must rely on advanced technologies and methodologies to extract oil from more challenging reservoirs. This requires a high level of operational efficiency and adaptability (Lu, Guo, Azimi, & Huang, 2019).

In recent years, Agile project execution frameworks have emerged as a potent solution to address oil production projects' dynamic and complex nature. Initially developed for software development, Agile methodologies emphasize iterative progress, flexibility, and close collaboration among project stakeholders (Alyatama, 2021). These principles can be effectively applied to the oil production sector to enhance project management and execution. Agile frameworks enable teams to respond swiftly to changes, reduce project risks, and improve overall project outcomes by fostering a culture of continuous improvement and adaptability (Gómez, 2022). The primary objective of this paper is to explore the application of Agile project execution frameworks in optimizing oil production processes amidst the complex challenges of the energy sector. This exploration will include an analysis of the specific challenges faced by the oil production industry, the benefits and principles of Agile methodologies, and strategies for implementing Agile frameworks in oil production projects. By synthesizing insights from various sources, this paper aims to comprehensively understand how Agile methodologies can transform oil production processes, leading to enhanced efficiency, safety, and profitability.

The scope of this paper encompasses a detailed examination of the complexities inherent in oil production, including technical, regulatory, and market-related challenges. It will delve into the principles and advantages of Agile frameworks, drawing comparisons with traditional project management approaches to highlight the unique benefits of Agile methodologies. Additionally, the paper will outline practical strategies for integrating Agile frameworks into oil production processes, identifying best practices, potential obstacles, and solutions to overcome them. Finally, the paper will offer conclusions and recommendations for industry stakeholders, suggesting future directions for research and development to further optimize oil production through Agile methodologies.

2. Challenges in the energy sector

2.1 Identification and analysis of key challenges in oil production

One of the foremost challenges in oil production is the fluctuating nature of global oil prices. Market volatility can be attributed to various factors, including geopolitical instability, changes in supply and demand, and economic cycles. This unpredictability makes it difficult for companies to plan long-term investments and manage budgets effectively, leading to potential financial strain (Khan, Su, Tao, & Umar, 2021).

Another critical challenge is the technical complexity involved in extracting oil from increasingly difficult environments. As easily accessible oil reserves become depleted, companies are forced to explore and develop reserves located in deep-water offshore fields, arctic regions, and unconventional sources such as shale formations (Henderson & Loe, 2016). These environments pose significant technical hurdles, requiring advanced technologies and expertise to ensure safe and efficient extraction. The high costs associated with developing such technologies and the risks involved further complicate the production process.

Environmental and regulatory pressures also present substantial challenges. The oil industry is subject to rigorous environmental regulations to reduce emissions, prevent spills, and protect ecosystems (Ite, Ufot, Ite, Isaac, & Ibok, 2016). Compliance with these regulations often requires substantial investments in new technologies and processes, which can increase operational costs. Additionally, there is growing public and governmental pressure to transition towards renewable energy sources, which places additional strain on the oil industry to innovate and improve its sustainability practices (Fattouh, Poudineh, & West, 2019). Operational inefficiencies stemming from aging infrastructure and equipment are another significant challenge. Much of the infrastructure used in oil production, such as pipelines, rigs, and refineries, is decades old and prone to failures (Akashraj & Maleith, 2020). These failures can lead to production downtime, increased maintenance costs, and safety hazards. Modernizing this infrastructure requires substantial capital investment, which can be a barrier for many companies (Lu *et al*, 2019).

The shortage of skilled labor is also a pressing issue. The oil industry requires a highly skilled workforce to manage complex production processes and technologies. However, a growing skills gap exists as experienced professionals retire and fewer young workers enter the industry. This shortage can lead to operational inefficiencies, as companies may struggle to find and retain the talent needed to maintain and enhance production processes (Tan & Tang, 2016).

2.2 Impact of these challenges on operational efficiency and productivity

The challenges identified above profoundly impact the operational efficiency and productivity of oil production companies. Market volatility and fluctuating oil prices can lead to financial instability, forcing companies to cut costs, delay projects, or even halt production in extreme cases (Sedighi, Mohammadi, Farahani Fard, & Sedighi, 2019). This uncertainty undermines the ability to make strategic long-term investments in technology and infrastructure, which are crucial for maintaining and improving production efficiency.

Technical complexities associated with extracting oil from challenging environments can lead to increased operational costs and project delays. The need for specialized equipment and expertise drives up the cost of production, while the inherent risks involved can result in accidents and environmental incidents, further impacting productivity and reputation (Epelle & Gerogiorgis, 2020). Environmental and regulatory pressures necessitate ongoing investments in compliance and sustainability measures. While these investments are crucial for minimizing environmental impact and meeting legal requirements, they also add to operational costs and complexity. Failure to comply with regulations can result in significant fines and legal repercussions, further straining resources (Falcone, 2020).

Aging infrastructure and equipment contribute to frequent breakdowns and maintenance issues, causing unplanned

production stoppages and safety concerns. These interruptions reduce overall productivity and increase the cost of repairs and replacements. Moreover, outdated infrastructure may not be compatible with modern technologies, limiting the ability to implement efficiency-enhancing innovations (Chin, Varbanov, Klemeš, Benjamin, & Tan, 2020).

The shortage of skilled labor exacerbates these challenges by making it difficult to maintain and optimize production processes. A lack of experienced personnel can lead to operational errors, reduced innovation, and an inability to implement new technologies and methodologies effectively. This skills gap can also slow the adoption of advanced practices necessary to address the industry's evolving challenges (Ambrogio, Filice, Longo, & Padovano, 2022).

2.3 The need for innovative approaches to address these challenges

Given the multitude of challenges faced by the oil production industry, there is a pressing need for innovative approaches that can enhance operational efficiency and productivity. One such approach is the adoption of Agile project execution frameworks (Post & Altman, 2017). Agile methodologies emphasize flexibility, collaboration, and iterative progress and can significantly benefit managing complex oil production projects. By fostering a culture of continuous improvement and adaptability, Agile frameworks enable companies to respond swiftly to changes in market conditions, technological advancements, and regulatory requirements (Noguera, Guerrero-Roldán, & Masó, 2018). Technology innovation is also critical. Developing and deploying advanced technologies such as automation, artificial intelligence, and data analytics can help optimize production processes, reduce operational costs, and enhance safety (Mithas, Chen, Saldanha, & De Oliveira Silveira, 2022). For instance, AI-powered predictive maintenance can anticipate equipment failures before they occur, minimizing downtime and maintenance costs. Similarly, data analytics can provide valuable insights into production performance, enabling companies to make informed decisions and optimize resource allocation (Javaid, Haleem, Singh, & Suman, 2022). Collaboration across the industry is another key to addressing these challenges. The industry can develop more effective solutions and share best practices by fostering partnerships between oil companies, technology providers, and regulatory bodies. Collaborative efforts can drive innovation, streamline regulatory compliance, and accelerate the adoption of new technologies and methodologies (Gray & Purdy, 2018). Investment in workforce development is essential to bridging the skills gap. Companies must prioritize training and development programs to equip their employees with the necessary skills to manage modern production technologies and processes. Encouraging young talent to enter the industry through educational partnerships and apprenticeship programs can also help build a robust and skilled workforce for the future (Mgonja, 2017).

3. Agile frameworks in project execution

3.1 Agile principles and methodologies

Agile principles emphasize iterative development, customer collaboration, and responsiveness to change. At the heart of Agile is the idea of breaking down projects into smaller,

manageable increments known as iterations or sprints. Each sprint typically lasts two to four weeks and involves crossfunctional teams working on specific deliverables. This iterative process allows for regular assessment and adaptation, ensuring that the project remains aligned with stakeholder needs and can quickly adapt to any changes or challenges (Shore & Warden, 2021).

Key methodologies within the Agile framework include Scrum, Kanban, and Extreme Programming (XP). Scrum is the most widely used Agile methodology and focuses on structured workflows and defined roles such as the Scrum Master, Product Owner, and development team. It involves regular ceremonies such as daily stand-ups, sprint planning, sprint reviews, and retrospectives to ensure continuous progress and improvement (Saleh, Huq, & Rahman, 2019). On the other hand, Kanban emphasizes visualizing workflow and limiting work in progress to optimize efficiency and identify bottlenecks. XP focuses on technical practices and improving software quality through techniques such as pair programming, test-driven development, and continuous integration (Hofmann, Lauber, Haefner, & Lanza, 2018).

The core values of Agile, as outlined in the Agile Manifesto, include prioritizing individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan. These values underscore the importance of flexibility, communication, and iterative progress in achieving project success (Alahyari, Svensson, & Gorschek, 2017).

3.2 Benefits of applying agile frameworks to oil production projects

Applying Agile frameworks to oil production projects offers numerous benefits that can significantly enhance project outcomes (Goyal, 2021). One of the primary advantages is increased flexibility and adaptability. The oil production industry is inherently complex and uncertain, including fluctuating market conditions, regulatory changes, and technical challenges. Agile methodologies enable teams to respond swiftly to these changes, adjust project plans, and reallocate resources as needed, thereby minimizing disruptions and maintaining project momentum (Ciccarelli, McLachlan, Singh, & Thomson, 2018).

Another significant benefit is improved collaboration and communication. Agile frameworks promote close interaction among cross-functional teams, stakeholders, and customers (Aslam, Saleem, Khan, & Kim, 2022). Regular meetings and reviews ensure that all parties are aligned on project goals and progress, fostering a collaborative environment where issues can be promptly identified and addressed. This enhanced communication reduces misunderstandings, accelerates decision-making, and ensures that the project remains on track (Aliasser & Adesta, 2021).

Agile methodologies also facilitate continuous improvement and innovation. By breaking down projects into smaller increments and conducting regular retrospectives, teams can constantly evaluate their performance, identify areas for improvement, and implement changes. This iterative process encourages experimentation and learning, leading to more innovative solutions and better project outcomes (Joshi, 2021). Moreover, Agile frameworks enhance transparency and accountability. Visual management tools, such as Kanban boards and burndown charts, provide clear visibility

into project progress and resource utilization. This transparency helps stakeholders understand the project's current status, anticipate potential issues, and make informed decisions. Additionally, the structured roles and responsibilities within Agile teams ensure that each team member is accountable for their tasks, promoting a sense of ownership and commitment (Williams, 2015).

3.3 Comparison with traditional project management approaches

Traditional project management approaches, often called Waterfall methodologies, follow a linear and sequential process. Projects are divided into distinct phases such as planning, execution, and closure, with each phase requiring completion before moving on to the next. This approach relies heavily on detailed upfront planning and predefined deliverables, with limited flexibility to accommodate changes once the project is underway (Reaiche & Papavasiliou, 2022).

One of the main differences between Agile and traditional approaches is the handling of change. In traditional project management, changes to the project scope, timeline, or budget are often viewed as disruptions that can jeopardize the project's success (Ciric Lalic, Lalic, Delić, Gracanin, & Stefanovic, 2022). Consequently, traditional methodologies emphasize strict adherence to the initial plan and rigorous change control processes. In contrast, Agile methodologies embrace change and view it as an opportunity to enhance the project's value. By incorporating regular feedback loops and iterative development, Agile frameworks can adapt to changing requirements and priorities, ensuring that the final deliverables meet stakeholder needs (Grebic, 2019). Another key distinction lies in the approach to project planning and execution. Traditional methodologies typically involve extensive upfront planning, with detailed specifications and schedules developed before any work begins. This can lead to significant time spent on documentation and planning, potentially delaying the start of actual project work. Agile methodologies, on the other hand, prioritize early delivery of working components and incremental progress. This allows for faster project initiation and the ability to deliver value to stakeholders early and often (Ahimbisibwe, Cavana, & Daellenbach, 2015). Collaboration and communication also differ significantly between the two approaches. Traditional project management often relies on hierarchical structures and formal communication channels, which can result in siloed teams and delayed decision-making. Agile frameworks promote a flat organizational structure with frequent and open communication among all team members and stakeholders. This collaborative environment facilitates faster problemsolving and ensures that everyone is aligned on project goals and progress (Anders, 2016).

Regarding risk management, traditional methodologies tend to front-load risk assessment and mitigation planning, focusing on identifying and addressing potential risks early in the project lifecycle. While this proactive approach can be effective, it may not fully account for complex projects' dynamic and unpredictable nature (Houston, 2019). Agile methodologies, by contrast, incorporate continuous risk assessment and mitigation throughout the project. Regular reviews and retrospectives enable teams to identify emerging risks and adjust their strategies accordingly, reducing the

likelihood of unexpected issues derailing the project (Shrivastava & Rathod, 2017).

In conclusion, Agile frameworks offer a compelling alternative to traditional project management approaches, particularly for complex and dynamic projects like oil production. By emphasizing flexibility, collaboration, continuous improvement, and transparency, Agile methodologies can help organizations navigate the complexities of oil production, enhance operational efficiency, and deliver better project outcomes. Embracing Agile principles and practices can lead to more resilient and adaptive project execution, ultimately driving greater success in the challenging and ever-evolving energy sector.

4. Implementing agile in oil production

4.1 Strategies for integrating agile frameworks into oil production processes

Integrating Agile frameworks into oil production requires a thoughtful and strategic approach tailored to the unique challenges and dynamics of the industry. One effective strategy is to start with pilot projects. By implementing Agile methodologies on a smaller scale, organizations can test their viability, gather insights, and refine practices before scaling up. Pilot projects also help build confidence and gain buy-in from stakeholders (Piya, Shamsuzzoha, Khadem, & AlHinai, 2020).

Another crucial strategy is to foster a cultural shift towards Agile values. This involves promoting a mindset that values collaboration, flexibility, and continuous improvement.

Leadership is key in driving this cultural change by modeling Agile behaviors, encouraging open communication, and supporting team autonomy. Training and development programs can also be instrumental in equipping employees with the necessary skills and knowledge to work effectively within Agile frameworks (Aslam *et al*, 2022).

Cross-functional teams are a cornerstone of Agile methodologies. In oil production, forming multidisciplinary teams that include geologists, engineers, project managers, and IT professionals can enhance collaboration and problemsolving. These teams should be empowered to make decisions and adjust plans as needed to respond to changing conditions and emerging challenges (Holdaway & Irving, 2017).

4.2 Best practices for successful implementation

Successful implementation of Agile in oil production hinges on several best practices. One such practice is maintaining a clear and shared vision. All team members and stakeholders should have a common understanding of project goals and desired outcomes. This alignment ensures that efforts are focused and coordinated, reducing the risk of miscommunication and conflicting priorities. Regular and effective communication is another critical practice. Agile methodologies emphasize frequent interactions, such as daily stand-ups, sprint reviews, and retrospectives. These meetings facilitate transparency, enable quick resolution of issues, and inform everyone about project progress. Additionally, leveraging digital communication tools can enhance connectivity, especially in geographically dispersed teams.

Incremental and iterative development is a fundamental Agile principle. Breaking down projects into smaller,

manageable segments allows for regular assessment and adaptation. This approach improves flexibility and enables early delivery of value to stakeholders. In oil production, this could translate to completing specific phases of exploration or drilling before moving on to the next, thereby mitigating risks and learning from each stage (Epelle & Gerogiorgis, 2020). A focus on continuous improvement is vital for Agile success. Retrospectives allow teams to reflect on their performance, identify areas for improvement, and implement changes. This practice fosters a culture of learning and adaptation, driving ongoing enhancements in efficiency and effectiveness (Moran, 2015).

4.3 Potential obstacles and solutions for overcoming them
Implementing Agile in oil production is not without challenges. One common obstacle is resistance to change. The transition to Agile methodologies can be daunting, especially for organizations with deeply entrenched traditional practices. To overcome this resistance, it is essential to communicate the benefits of Agile clearly and demonstrate success through pilot projects. Engaging stakeholders middle and involving them in the change process can also help secure their support (Prestidge, 2022). Another challenge is the alignment of Agile practices with regulatory requirements. The oil production industry is highly regulated, and ensuring compliance while maintaining Agile flexibility can be challenging. This can be addressed by integrating compliance checks into Agile processes, such as incorporating regulatory review points within sprints and maintaining close collaboration with legal and compliance teams.

Resource constraints can also pose a significant hurdle. Agile frameworks require dedicated, cross-functional teams, which may be difficult to assemble due to limited availability of skilled personnel. Addressing this challenge involves strategic resource planning and prioritization. Organizations may need to invest in training and development to build the necessary skills internally or consider external partnerships to fill gaps (Narayan, 2015).

4.4 Tools and technologies that support agile project execution in oil production

The use of specialized tools and technologies greatly facilitates the successful implementation of Agile in oil production. Project management tools such as JIRA, Trello, And Asana are essential for planning, tracking, and managing Agile projects. These tools provide features such as task boards, sprint planning, and progress tracking, which are crucial for maintaining transparency and accountability. Collaboration tools like Slack, Microsoft Teams, and Zoom enhance communication and coordination among team members. These platforms support real-time messaging, video conferencing, and file sharing, enabling seamless collaboration even in distributed teams. These tools are invaluable for oil production projects that often involve remote locations and field operations.

Data analytics and visualization tools are critical in Agile project execution. Tools such as Power BI, Tableau, and custom dashboards help track key performance indicators, monitor project health, and make data-driven decisions. In oil production, real-time data from drilling operations, equipment performance, and environmental conditions can be integrated into these tools to provide actionable insights

and facilitate rapid response to changing conditions (Vesterinen, 2021). Advanced simulation and modeling tools are particularly relevant in oil production. Software such as Petrel, Eclipse, and CMG are used for reservoir modeling, seismic interpretation, and production forecasting. Integrating these tools within Agile frameworks allows for iterative testing and refinement of models, enhancing accuracy and reducing uncertainty (Al-Bayati, 2017). Moreover, the adoption of cloud computing and digital twin technologies can significantly enhance Agile project execution. Cloud platforms such as AWS, Azure, and Google Cloud provide scalable data storage, processing, and analytics infrastructure. Digital twins, which are virtual replicas of physical assets, enable real-time monitoring, simulation, and optimization of oil production processes. These technologies support Agile principles by facilitating continuous feedback, real-time decision-making, and iterative improvement (Nath, Van Schalkwyk, & Isaacs, 2021).

5. Conclusion and recommendations

5.1 Conclusion

Optimizing processes in the ever-evolving and complex landscape of oil production is crucial for enhancing efficiency and maintaining competitive advantage. This paper has explored the integration of Agile frameworks into oil production, emphasizing its potential to address inherent challenges and improve operational outcomes. Key findings underscore Agile methodologies' significant benefits, such as enhanced flexibility, improved team collaboration, and increased adaptability to changing conditions. By adopting Agile principles, oil production projects can transition from rigid, traditional management practices to more dynamic and responsive approaches, leading to better alignment with industry demands and market fluctuations.

Despite its traditionalist leanings, one of the main insights is that the oil production sector can greatly benefit from Agile's iterative and incremental development processes. These processes allow for continuous improvement and rapid response to unexpected challenges, such as fluctuating market prices, regulatory changes, and technological advancements. The implementation of cross-functional teams, empowered to make decisions and adapt quickly, is another crucial aspect of successful Agile integration. These teams foster a collaborative environment encouraging innovation and efficient problem-solving, ultimately driving project success.

Moreover, using advanced tools and technologies is critical in supporting Agile project execution. Project management software, collaboration platforms, data analytics, and simulation tools are vital in facilitating Agile practices. These tools enhance communication and coordination among team members and provide real-time data and insights that are essential for informed decision-making.

5.2 Recommendations

For industry stakeholders considering the adoption of Agile frameworks in oil production, several recommendations can help ensure a smooth and successful transition. Firstly, it is essential to start small and scale gradually. Implementing Agile methodologies in pilot projects allows organizations to test and refine their approach before broader application. This incremental implementation helps build confidence and

gain stakeholder buy-in, which is critical for long-term success.

Stakeholders should also focus on fostering a cultural shift towards Agile values. This involves promoting an organizational culture that values flexibility, collaboration, and continuous improvement. Leadership must play an active role in modeling Agile behaviors and supporting teams through training and development initiatives. Encouraging open communication and transparency is also vital, as it builds trust and ensures that all team members are aligned with project goals and expectations.

Investing in the right tools and technologies is another key recommendation. Stakeholders should leverage project management and collaboration tools that support Agile practices and data analytics and simulation software that provide real-time insights. These tools facilitate Agile workflows and enhance the overall efficiency and effectiveness of project execution.

Addressing potential obstacles proactively is crucial for successful Agile adoption. Resistance to change is a common barrier, and it can be mitigated by clearly communicating the benefits of Agile methodologies and demonstrating success through pilot projects. Ensuring compliance with regulatory requirements while maintaining Agile flexibility can be challenging but is achievable through integrated compliance checks and close collaboration with legal teams. Finally, stakeholders should prioritize continuous learning and improvement. Regular retrospectives and feedback loops are essential components of Agile frameworks, enabling teams to reflect on their performance, identify areas for improvement, and implement changes. This iterative process fosters a culture of learning and adaptation, driving ongoing enhancements in project execution and overall organizational performance.

6. References

1. Ahimbisibwe A, Cavana RY, Daellenbach U. A contingency fit model of critical success factors for software development projects: A comparison of agile and traditional plan-based methodologies. *Journal of Enterprise Information Management*. 2015;28(1):7–33.
2. Akashraj D, Maleith KD. The impact of ageing facilities on oil production in South Sudan. *International Journal of Research and Review*. 2020;7.
3. Al-Bayati F. Performance of the horizontal wells in a naturally fractured carbonate reservoir. [Dissertation]. 2017.
4. Alahyari H, Svensson RB, Gorschek T. A study of value in agile software development organizations. *Journal of Systems and Software*. 2017; 125:271–88.
5. Aliasser A, Adesta EY. Development of agile project management framework in oil and gas companies in Kuwait. *Asian Journal of Electrical and Electronic Engineering*. 2021;1(1):8–14.
6. Alyatama M. Application of agile project management in Kuwait oil and gas capital projects. [Dissertation]. University of Northampton. 2021.
7. Ambrogio G, Filice L, Longo F, Padovano A. Workforce and supply chain disruption as a digital and technological innovation opportunity for resilient manufacturing systems in the COVID-19 pandemic. *Computers & Industrial Engineering*. 2022; 169:108158.
8. Anders A. Team communication platforms and emergent social collaboration practices. *International Journal of Business Communication*. 2016;53(2):224–61.
9. Arya AK. A critical review on optimization parameters and techniques for gas pipeline operation profitability. *Journal of Petroleum Exploration and Production Technology*. 2022;12(11):3033–57.
10. Aslam J, Saleem A, Khan NT, Kim YB. Blockchain technology for oil and gas: implications and adoption framework using agile and lean supply chains. *Processes*. 2022;10(12):2687.
11. Chin HH, Varbanov PS, Klemeš JJ, Benjamin MFD, Tan RR. Asset maintenance optimisation approaches in the chemical and process industries: A review. *Chemical Engineering Research and Design*. 2020; 164:162–94.
12. Ciccarelli J, McLachlan D, Singh H, Thomson R. Agile field development planning: A systems approach. Paper presented at: SPE Asia Pacific Oil and Gas Conference and Exhibition. 2018.
13. Ciric Lalic D, Lalic B, Delić M, Gracanin D, Stefanovic D. How project management approach impacts project success? From traditional to agile. *International Journal of Managing Projects in Business*. 2022;15(3):494–521.
14. Epelle EI, Gerogiorgis DI. A review of technological advances and open challenges for oil and gas drilling systems engineering. *AIChE Journal*. 2020;66(4):e16842.
15. Falcone PM. Environmental regulation and green investments: The role of green finance. *International Journal of Green Economics*. 2020;14(2):159–73.
16. Fattouh B, Poudineh R, West R. The rise of renewables and energy transition: What adaptation strategy exists for oil companies and oil-exporting countries? *Energy Transitions*. 2019;3(1):45–58.
17. Gielen D, Gorini R, Wagner N, Leme R, Gutierrez L, Prakash G, *et al* Global energy transformation: A roadmap to 2050. IRENA. 2019.
18. Gómez EDS. Scalable agile frameworks in large enterprise project portfolio management. [Dissertation]. Pontificia Universidad Católica del Perú. 2022.
19. Goyal A. Enhancing engineering project efficiency through cross-functional collaboration and IoT integration. *International Journal of Research and Analytical Reviews*. 2021;8(4):396–402.
20. Gray B, Purdy J. Collaborating for our future: Multistakeholder partnerships for solving complex problems. Oxford University Press. 2018.
21. Grebić B. Traditional vs agile project management in the service sector. *European Project Management Journal*. 2019;9(2):55–63.
22. Henderson J, Loe JSP. The prospects and challenges for Arctic oil development. [Report]. 2016.
23. Hofmann C, Lauber S, Haefner B, Lanza G. Development of an agile development method based on Kanban for distributed part-time teams and an introduction framework. *Procedia Manufacturing*. 2018; 23:45–50.
24. Holdaway KR, Irving DH. Enhance oil and gas exploration with data-driven geophysical and petrophysical models. John Wiley & Sons. 2017.

25. Houston DA. Knowledge management and positive deviance: A study of construction project outcomes. [Dissertation]. Capella University. 2019.
26. Ite AE, Ufot UF, Ite MU, Isaac IO, Ibok UJ. Petroleum industry in Nigeria: Environmental issues, international environmental legislation and implementation of international environmental law. *American Journal of Environmental Protection*. 2016;4(1):21–37.
27. Javaid M, Haleem A, Singh RP, Suman R. Artificial intelligence applications for Industry 4.0: A literaturebased study. *Journal of Industrial Integration and Management*. 2022;7(01):83–111.
28. Joshi PL. A review of agile internal auditing: Retrospective and prospective. *International Journal of Smart Business and Technology*. 2021;9(2):13–32.
29. Khan K, Su CW, Tao R, Umar M. How do geopolitical risks affect oil prices and freight rates? *Ocean & Coastal Management*. 2021; 215:105955.
30. Lu H, Guo L, Azimi M, Huang K. Oil and Gas 4.0 era: A systematic review and outlook. *Computers in Industry*. 2019; 111:68–90.
31. Mgonja C. Enhancing the university-industry collaboration in developing countries through best practices. *International Journal of Engineering Trends and Technology*. 2017;50(4):216–25.
32. Mithas S, Chen ZL, Saldanha TJ, De Oliveira Silveira A. How will artificial intelligence and Industry 4.0 emerging technologies transform operations management? *Production and Operations Management*. 2022;31(12):4475–87.
33. Moran A. *Managing agile: Strategy, implementation, organisation, and people*. Cham: Springer. 2015.
34. Narayan S. *Agile IT organization design: For digital transformation and continuous delivery*. AddisonWesley Professional. 2015.
35. Nath SV, Van Schalkwyk P, Isaacs D. *Building industrial digital twins: Design, develop, and deploy digital twin solutions for real-world industries using Azure digital twins*. Packt Publishing Ltd. 2021.
36. Noguera I, Guerrero-Roldán AE, Masó R. Collaborative agile learning in online environments: Strategies for improving team regulation and project management. *Computers & Education*. 2018; 116:110–29.
37. Pelluru K. Unveiling the power of IT DataOps: Transforming businesses across industries. *Innovative Computer Sciences Journal*. 2022;8(1):1–10.
38. Piya S, Shamsuzzoha A, Khadem M, Al-Hinai N. Identification of critical factors and their interrelationships to design agile supply chain: Special focus to oil and gas industries. *Global Journal of Flexible Systems Management*. 2020; 21:263–81.
39. Post JE, Altman BW. Managing the environmental change process: Barriers and opportunities. In: *Managing Green Teams*. Routledge; 2017. p. 84–101.
40. Prestidge KL. Digital transformation in the oil and gas industry: Challenges and potential solutions. [Dissertation]. Massachusetts Institute of Technology. 2022.
41. Reaiche C, Papavasiliou S. The traditional, sequential methodologies. In: *Management Methods for Complex Projects*. 2022.
42. Saleh SM, Huq SM, Rahman MA. Comparative study within Scrum, Kanban, XP focused on their practices. Paper presented at: 2019 International Conference on Electrical, Computer and Communication Engineering (ECCE). 2019.
43. Sedighi M, Mohammadi M, Farahani Fard S, Sedighi M. The nexus between stock returns of oil companies and oil price fluctuations after heavy oil upgrading: Toward theoretical progress. *Economies*. 2019;7(3):71.
44. Shore J, Warden S. *The art of agile development*. O'Reilly Media, Inc.; 2021.
45. Shrivastava SV, Rathod U. A risk management framework for distributed agile projects. *Information and Software Technology*. 2017; 85:1–15.
46. Tan KS, Tang JT. New skills at work: Managing skills challenges in ASEAN-5. [Report]. 2016.
47. Vesterinen S. Creation of automated EazyBI data graphs for project progress reporting. [Dissertation]. 2021.
48. Williams P. *Visual project management*. Lulu.com. 2015.